

Reshma Rajendran

Machine Learning Engineer | 2+ years ML experience (audio, computer vision, NLP) | MTech IIT Tirupati

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Machine Learning Engineer with experience in signal processing, computer vision, machine learning and deep learning. Skilled in MATLAB, Python with hands-on use of NumPy, Sklearn, and OpenCV. Proficient in image processing, computer vision techniques and deep learning frameworks (PyTorch). Familiar with Computer vision applications such as image classification, object detection and anomaly detection.

EXPERIENCE

Aiverbalyze technologies | Machine learning Engineer

May 2025–present | India

Fine-tuned a Llama based TTS model, to generate highly expressive, natural-sounding speech in multiple Indian languages.

- Engineered a “The voice” production-grade multilingual TTS system trained on 2.2M+ audio-text pairs, achieving low training loss of 3.43 and ~500ms streaming latency, ensuring seamless interaction with the TTS system.
- Optimized training with Unsloth (LoRA) and Axolotl, reducing compute cost, training time, and memory footprint by 50% while maintaining model quality.

Abili-Tech System | Machine learning Engineer

May 2024–April 2025 | BC, Canada

Developed a real-time sign language translation web app using Streamlit, deployed on Hugging Face Spaces.

- Implemented a Random Forest model achieving ~75% translation accuracy for gesture-to-text conversion.
- Used OpenCV for real-time video capture and optimized data preprocessing, improving responsiveness and reliability.

NXP Semiconductors | Machine learning Intern

Jun 2022–Jun 2023 | Bangalore, India

Collaborated within NXP Semiconductors' R&D team, gaining expertise in audio classification and signal analysis.

- Designed and implemented audio detection system with specific audio features leveraging dominant frequency and high SNR, achieving 22% Equal Error Rate over baseline features with lower computational complexity.
- Applied Gaussian Mixture Model (GMM) for multi-class audio inference, enhanced feature quality with custom filter bank and spectrum thresholding and proposed temporal variation analysis for future improvements.

Tata Consultancy Services | Assistant System Engineer

Aug 2017–April 2019 | Chennai, India

- Packaged, tested, and deployed applications via SCCM during a Windows 10 migration for 10,000+ users
- Managed Hyper-V hosts and internal lab infrastructure for demos and testing, while developing PowerShell and Azure automation scripts to streamline SCCM tasks, reducing manual effort and improving operational efficiency.

PROJECTS

Anomaly Detection on MVTec Dataset – Explored multiple approaches for anomaly detection including HOG+SVM and autoencoders, which underperformed on certain categories (AUC ~0.36–0.74). Developed a ResNet-based feature extraction pipeline with a neural network classifier, achieving a significant performance boost to 0.86 AUC.

Image classification: Implemented a custom Convolutional Neural Network (CNN) from scratch using PyTorch and trained it on the 10 Monkey Species dataset, achieving a training accuracy of 92.34% and validation accuracy of 83.46%.

SKILLS

Languages/libraries: Python (pandas, NumPy, Scikit-Learn, NLTK, OpenCV, PyTorch), MATLAB, SQL

Computer Vision: Image preprocessing, feature extraction, classification, segmentation, object detection

NLP: Text preprocessing, probabilistic models, sequence models, attention models, transformers

EDUCATION

Indian Institute of Technology (IIT) Tirupati

Aug 2021–May 2023 | Tirupati, India

Master of Technology (MTech) in Electrical Engineering| GPA: 8.73/10

- Specialization: Signal Processing and Communication
- Relevant Courses: Deep Learning for Computer Vision, Machine Learning for Image Processing

Rajalakshmi Engineering College

Aug 2013–May 2017 | Chennai, India

Bachelor of Engineering (BE) in Electronics and Communication Engineering| GPA: 8.46/10

CERTIFICATIONS

Deep Learning online course | One Fourth LABS (IIT MADRAS)

Mastering OpenCV with Python | OpenCV University

AWS Certified Cloud Practitioner | AWS